



SHIVNETRA 47

SHIV-NETRA47 AIR DEFENCE SYSTEM PRIVATE LIMITED

AUTONOMOUS AIR SYSTEMS • SWARM INTELLIGENCE • TACTICAL AIR DEFENCE

🇮🇳 MAKE IN INDIA • ATMANIRBHAR BHARAT

Engineering Collaboration: **BCN In-House Technology Team**

▬ DECENTRALIZED SWARM COMMAND & BATTLESPACE DYNAMICS

HETEROGENEOUS SWARM FLEET & AUTONOMOUS MESH C2

A resilient, self-healing peer-to-peer RF mesh architecture coordinating disposable sub-500g Nano-scouts and long-endurance Overwatch VTOL relays—delivering continuous multi-vector border overwatch, collaborative targeting, and uncompromised survivability under electronic warfare denial.

● SQUAD (6 UNITS) & THEATRE (24+ UNITS)

◆ PEER-TO-PEER ENCRYPTED RF MESH

▲ BVLOS 100+ KM COVERAGE

★ ATTRITION-TOLERANT SELF-HEALING

6-NODE TACTICAL SQUAD SPREAD

SQUAD RECON • RAPID FORWARD POST



ARCH-REF: SN47-SYS-06

FORWARD BORDER PATROL & PERIMETER BREACH DETECTION

Lightweight tactical deployment engineered for company-level outposts and rapid reconnaissance. Couples four agile micro Nano-scouts with two high-altitude Overwatch relays for continuous localized overwatch.

FLEET COMPOSITION

4x Nano + 2x Overwatch

INTRA-SWARM LATENCY

<85 ms Real-time

TACTICAL MESH RANGE

25-40 km Line-of-Sight

SETUP & DEPLOYMENT

<4 Minutes

- Low-acoustic forward scout leapfrogging across steep mountain ridges
- Dual airborne relay redundancy ensuring unbroken squad tablet video feed
- Autonomous terrain-following avoiding forward radar cross-sections

01 P2P ENCRYPTED MESH

Ad-hoc decentralized peer-to-peer data-links operating on frequency-hopping spread spectrum (FHSS). Every drone functions as an autonomous transceiver relay.

02 BOUNDING OVERWATCH

Automated leapfrogging logic: forward Nano-scouts advance while high-altitude Overwatch relays maintain elevated optical overwatch and RF line-of-sight.

6 to 64+ Nodes

SWARM MESH SCALABILITY

Modular protocol expansion from squad reconnaissance to brigade-level saturation.

<45 ms

INTRA-SWARM ROUTING LATENCY

Low-latency tactical telemetry and telemetry relay across multi-hop peer nodes.

24+ NODE SATURATION THEATRE SCREEN

THEATRE OVERWATCH • HIGH DENSITY



ARCH-REF: SN47-SYS-24

WIDE-AREA CONTESTED INFILTRATION CURTAIN & C-UAS SCREEN

Wide-area sovereign border curtain deploying 18+ Nano-scouts coordinated by 6 multi-spectral Overwatch relays. Delivers continuous perimeter sealing and collaborative target tracking across complex terrain.

FLEET COMPOSITION

18x Nano + 6x Overwatch

SIMULTANEOUS TRACKS

32+ Target Vectors

OPERATIONAL COVERAGE

100+ km Sector Radius

ATTRITION SURVIVABILITY

Up to 40% Node Loss

- Decentralized dynamic multi-hop packet routing without single point of failure
- Collaborative bounding overwatch and automatic target handoff between nodes
- Self-reconstituting mesh geometry filling coverage holes if nodes are engaged

03 DYNAMIC TARGET HANDOFF

Continuous kinematic track maintenance: when targets traverse boundaries, adjacent swarm nodes automatically reposition and slew optical gimbals to preserve lock.

04 GPS-DENIED AUTONOMY

Under aggressive electronic jamming or GNSS denial, swarm units switch to visual inertial odometry (VIO) and NAVIC backup to maintain spatial precision.

32 dB

ANTI-JAM PROCESSING GAIN

Frequency-hopping spread spectrum with robust immunity in dense electronic countermeasure zones.

<250 ms

DYNAMIC RECONSTITUTION

Instantaneous network re-routing and autonomous sector coverage filling upon node attrition.